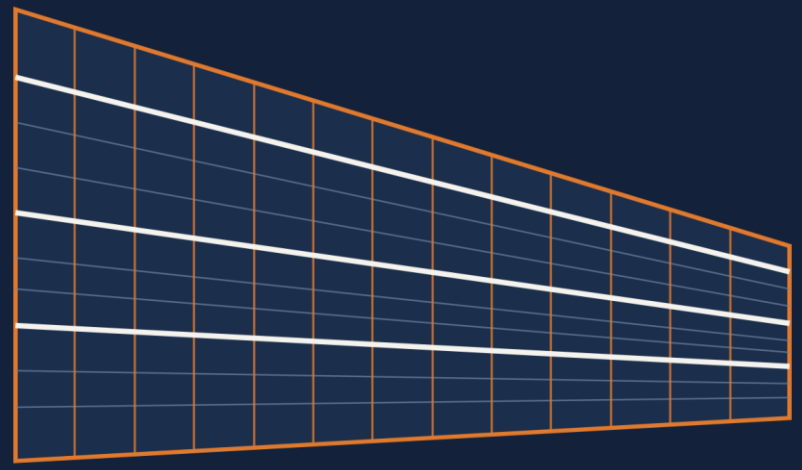
PROJECT PROPOSAL · 30 SEPTEMBER 2026

Parametric APDL-Based Horizontal Tailplane Structural Analysis & Optimization



A script-driven finite element workflow for sizing a lighter tailplane — adapted from Palumbo et al. (2026), HERFUSE project

ANSYS Mechanical APDL · MATLAB

MOTIVATION

Why a parametric tailplane model

In a manual FE model, every design change means a rebuild — rebuild — so only a few layouts layouts ever get tested.

Our answer: one script where every dimension is a variable.

A critical component

The HTP carries pitch stability and control loads — it must be strong, light and manufacturable.

New loading environments

Hybrid-electric and hydrogen aircraft change masses and load cases, so structures are re-sized often.

The bottleneck

Hand-built models are slow to modify, which limits how many designs we can compare early on.

LABORATORY AIM

Five objectives

01 **Parametric APDL model**
Geometry, mesh, materials and boundary conditions generated from span, chord, rib/spar/stringer counts and skin thickness.

02 **Load pre-processing**
Convert internal shear forces and bending moments into external nodal loads using a least-norm solution.

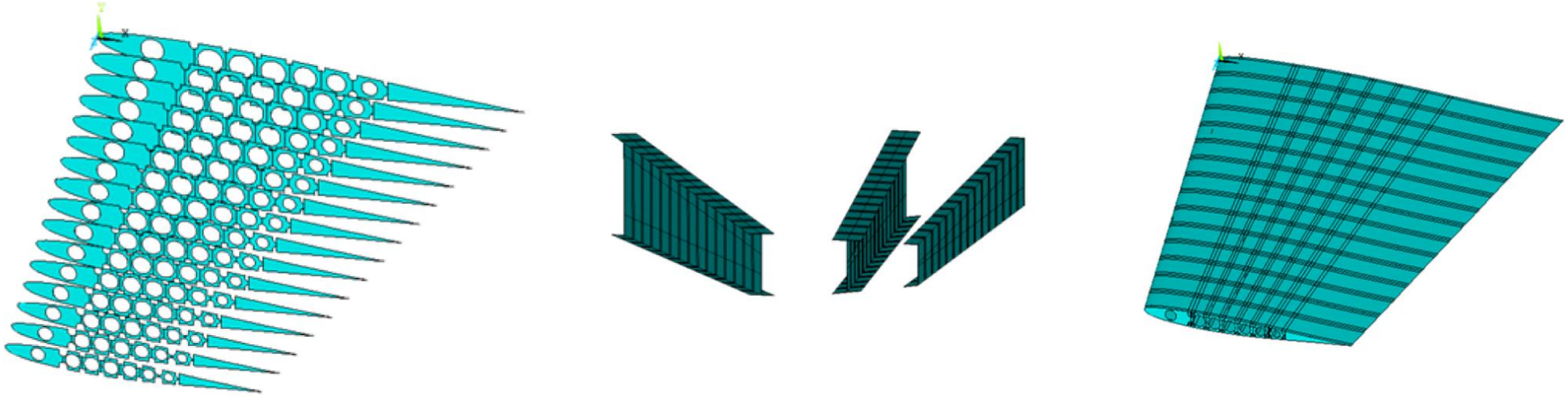
03 **Configuration study**
Compare HTP layouts to minimise mass while meeting strength and stiffness requirements.

04 **Model verification**
Free-free modal, unit-displacement and 1 g gravity checks before any aerodynamic load is applied.

05 **Reusable workflow**
A new airfoil, material or load case becomes a parameter update, not a model rebuild.

RELATED THEORY · FEM FOR THIN-WALLED STRUCTURES

How the tailplane is idealised



Ribs · SHELL181
Lightening holes made by cut operations

Spars · SHELL181
Shear webs; count set by N_SPAR

Skins · SHELL181
Equivalent isotropic properties from the laminate

Stringers · BEAM188
Timoshenko beam, parametric cross-section

Figure: (a) ribs · (b) spars · (c) full assembly and mesh — Palumbo et al. (2026)

STAGE 2A · PLANNED SCRIPT LOGIC

One script, every variable a parameter

</> In development

1
Parameters
Span, chords, N_RIB, N_SPAR, N_STR, T_SKIN, ESIZE

→

2
Stations
*DO loop places each rib along the span

→

3
Geometry
Rib and spar areas, skin surfaces, stringer lines

→

4
Lightening holes
Cut operations on each rib

5
Properties
Element types, materials, sections

→

6
Mesh
SHELL181 / BEAM188, one ESIZE

→

7
Supports & loads
Fix the HTP–VTP root, read the load file

→

8
Solve & extract
Mass, max σ_VM, tip deflection

↺ Change N_RIB (or any parameter) → steps 2–8 rerun by themselves. Configs A, B and C are just three parameter sets.

STAGE 2B · LOAD PRE-PROCESSING (OVERVIEW)

Turning sizing loads into FE loads

Step 3 · 1–2 Oct

INPUT

Internal sizing loads

Shear force and bending moment along the span (SFBM)

→

PROCESS · MATLAB

Load reconstruction

Find external forces that reproduce the same V and M (least-norm method, Palumbo et al.)

→

OUTPUT

APDL load file

Nodal forces applied directly to the FE model

Why this step is needed

An FE solver needs external forces on nodes, but tailplane sizing loads are usually given as internal shear and moment. This step bridges the two.

What we will study in Step 3

- CS-25 load cases and balancing tail load for the ATR 42-600
- The SFBM procedure in the reference paper
- Writing the result as an APDL load file

METHODOLOGY

A three-stage workflow

STAGE 1

Input data

Geometry, materials and SFBM sizing loads — all entered as APDL parameters

→

STAGE 2a

APDL script generation

Geometry, mesh, material cards

STAGE 2b · in parallel

Load pre-processing

MATLAB least-norm → APDL loads

→

STAGE 3

- **3a** Configuration setup · A, B, C
- **3b** Model verification
- **3c** Static analysis · σ_VM, tip deflection
- **3d** Evaluate · min mass, FS > 1.5

↺ **Feedback loop:** evaluation results set new parameter values → the script regenerates the model from Stage 2a

Workflow adapted from Palumbo et al. (2026)

STAGE 1 · REFERENCE AIRCRAFT

Why the ATR 42-600

The HERFUSE reference model is **confidential** — its geometry and loads are not published. We need an open aircraft of the same class, so every input is traceable.

1 · Same aircraft class

HERFUSE targets future hybrid-electric regional turboprops (led by Leonardo, co-owner of ATR). The ATR 42-600 is today's aircraft in that class.

2 · T-tail layout

The HTP sits on top of the fin, which matches the boundary condition in our model: fixed at the HTP–VTP root.

3 · Public, traceable data

MTOW and speed limits come from published type-certificate and manufacturer data, so reviewers can check every value.

4 · CS-25 load basis

Certified to CS-25: established load cases, including the balancing tail load, and a 1.5 factor of safety.

STAGE 1 · INPUT DATA

What goes into the script

Stage 1 complete

18,600 kg
MTOW

250 KIAS
V_C / V_MO

FS 1.5
Required factor of safety (CS-25)

HTP geometry → script parameters

Parameter	Script name	Value
Half-span	SPAN	[__ m]
Root chord	C_ROOT	[__ m]
Tip chord	C_TIP	[__ m]
Dihedral	DIHED	[__ °]
Thickness-to-chord	TC	[__ %]

Materials

Al 7075-T6 · ribs, spars, stringers	Value
Young's modulus E	71.7 GPa
Poisson's ratio ν	0.33
Density ρ	2,810 kg/m³
Yield strength σ_y	503 MPa
Composite skins	Equivalent isotropic

Sources: team parameter summary (ATR 42-400/600); typical handbook values for 7075-T6

STAGE 3A · CONFIGURATION SETUP

Three configurations, one script

Parameter	A · Baseline	B · Moderate	C · Aggressive
Ribs	14	9	7
Spars	3	2	2
Stringers	8	8 (unchanged)	6
Skin thickness	t0 (baseline)	t0 + Δ	[to be set]
Intent	Literature reference (HERFUSE)	Fewer members, thicker skin	Minimum mass that still passes

Selected design = **lowest structural mass** with **FS > 1.5** under the reconstructed SFBM loads

STAGE 3B–3D · VERIFICATION AND CRITERIA

Trust the model before loading it

Free-free modal

No supports applied.

Pass: exactly 6 modes ≈ 0 Hz before the first flexible mode. More zeros → a disconnected part.

Unit displacement

Rigid-body unit translation of all nodes.

Pass: stresses and reactions ≈ 0 → stiffness matrix assembled correctly.

1 g gravity

Gravity load, root fixed.

Pass: sum of root reactions = m · g of the model.

FS = σ_allowable / σ_VM,max > 1.5 Strength criterion for every configuration; mass and tip deflection reported alongside

PLAN

Schedule and budget

ID	Dates	Task	Status
T-01	16–22 Sep	Initial topic study (window structure)	Completed
T-02	23–25 Sep	Topic change to HTP; literature review	Completed
T-03	26–27 Sep	Step 1 · Data collection & parameters	Completed
T-04	28–30 Sep	Step 2 · Parametric APDL script	In progress
T-05	30 Sep	Proposal presentation (milestone)	Today
T-06	1–2 Oct	Step 3 · SFBM load pre-processing	Pending
T-07	3–4 Oct	Steps 4–5 · Config setup & verification	Pending
T-08	5–6 Oct	Steps 6–7 · Static & baseline validation	Pending
T-09	7–8 Oct	Step 8 · Evaluation & comparison	Pending
T-10	9–10 Oct	Step 9 · Final report preparation	Pending

BUDGET

0 THB

ANSYS Mechanical APDL 2026 R1 — Student version, free

MATLAB — university licence

Workstations — personal laptops

10 tasks · 3 completed · 1 in progress · 6 pending — Project Schedule Tracking

EXPECTED OUTCOMES

Deliverables, limits and next steps

Deliverables

- Parametric APDL script for rib, spar, stringer counts and skin thickness
- MATLAB SFBM load module
- Free-free modal and 1 g verification results
- Mass and FS comparison of A, B, C

Limitations

- No buckling (stability) check yet
- Equivalent isotropic skins, not ply-by-ply
- Simplified airfoil profile
- Student licence node limit caps mesh density

Future work

- Eigenvalue buckling analysis
- Composite layup definitions
- Aeroelastic coupling
- Automated optimiser / MDO loop

IN ONE LINE

Change a parameter — not the model.

With approval today, Steps 3–9 run from 1 to 10 October.

Thank you · Questions?